CH365 Chemical Engineering Thermodynamics

Lesson 31

Two-Phase Systems, Thermodynamic Diagrams, and Property Tables

Lessons 28-30 Recap

- Mathematics from Maxwell leads directly to Gibbs energy generating functions.
- Gibbs energy generating functions lead directly to residual properties G^R, H^R, and S^R and a recipe to calculate them.
- Residual properties G^R, H^R, and S^R add to ideal gas properties G^{ig}, H^{ig}, and S^{ig} to generate real properties G, H, and S.
- Generalized methods – Lee-Kesler Tables and Virial Equation that are somewhat simpler but not as accurate.

Today's Agenda

- Vapor-liquid phase equilibrium.
- Origin of the widely used Antoine equation:

$$\ln P^{sat} = A - \frac{B}{T + C} \qquad \text{(Eq. 6.76)}$$

A, B, and C are constants in Table B.2 on page 682

P^{sat} is vapor pressure in kPa and T is temperature in °C

Used in flash and distillation calculations – enables Raoult's Law calculations

Fundamental Property Relations

Property relations for a homogenous fluid of constant composition:

$dU = TdS - PdV$ (Eq.6.8)	$dH = TdS + VdP$ (Eq. 6.9)
$dA = -PdV - SdT$ (Eq. 6.10)	$dG = VdP - SdT$ (Eq. 6.11)

- Gibbs energy is the "driving force" for phase change (postulate).
- Consider a system with two phases, α and β.
- The system is closed and includes α and β.
- The system is at equilibrium.

$dG = 0$ for the system $\alpha + \beta$

$$dG^{\alpha} = dG^{\beta}$$

$$G^{\alpha} = G^{\beta} \quad \text{(Eq. 6.83)}$$

- Equilibrium is the absence of change.
- It follows that the Gibbs energy in each phase is equal.
- A *fundamental condition* for equilibrium is that $dG^{\alpha}=dG^{\beta}$

$$dH = dQ$$

$$dS = \frac{dQ}{T} = \frac{dH}{T}$$

$$dG = dH - TdS = dH - T\frac{dH}{T} = 0$$

A vapor initially in equilibrium with its liquid undergoes a differential amount of evaporation.

$$dG = VdP - SdT \quad \text{(Eq. 6.11)}$$

$$dG^{\alpha} = dG^{\beta}$$

$$V^{\alpha}dP - S^{\alpha}dT = V^{\beta}dP - S^{\beta}dT$$

For a differential amount of evaporation, dP and dT approach zero but are not zero

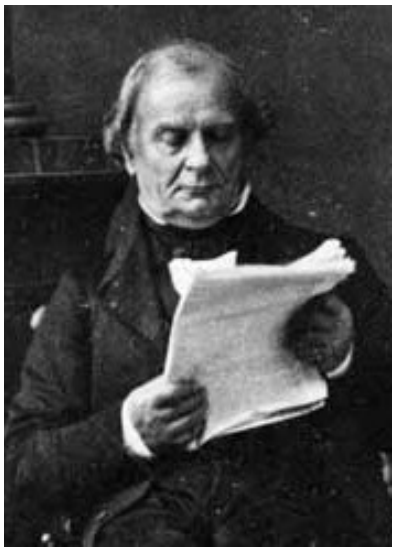
$$V^{\alpha}dP^{sat} - S^{\alpha}dT = V^{\beta}dP^{sat} - S^{\beta}dT$$

At equilibrium $P = P^{sat}$

$$V^{\alpha}dP^{sat} - V^{\beta}dP^{sat} = S^{\alpha}dT - S^{\beta}dT$$

$$\left(V^{\alpha} - V^{\beta}\right)dP^{sat} = \left(S^{\alpha} - S^{\beta}\right)dT$$

$$\frac{dP^{sat}}{dT} = \frac{S^{\alpha} - S^{\beta}}{V^{\alpha} - V^{\beta}} = \frac{\Delta S^{\alpha\beta}}{\Delta V^{\alpha\beta}}$$



Benoit Clapeyron, 1799-1864
(image from Wikipedia)

$$dH = TdS + VdP \quad \text{(Eq. 6.9)}$$

Consider for transition from α to β
Integrate at constant T and P
(pressure term goes away).

$$\Delta H^{\alpha\beta} = T\Delta S^{\alpha\beta} \quad \text{(Eq. 6.84)}$$

$$\Delta S^{\alpha\beta} = \frac{\Delta H^{\alpha\beta}}{T} \quad \text{(Eq. 6.84)}$$

$$\frac{dP^{sat}}{dT} = \frac{\Delta S^{\alpha\beta}}{\Delta V^{\alpha\beta}} = \frac{\Delta H^{\alpha\beta}}{T\Delta V^{\alpha\beta}} \quad \text{(Eq. 6.85)}$$

Clapeyron Equation
(basis for phase change enthalpy calculations)

Transition from liquid l to vapor v (change labels α and β to l and v)

$$\frac{dP^{sat}}{dT} = \frac{\Delta H^{\alpha\beta}}{T\Delta V^{\alpha\beta}} \Rightarrow \frac{dP^{sat}}{dT} = \frac{\Delta H^{lv}}{T\Delta V^{lv}} \quad \text{(Eq. 6.86)}$$

$$Z \equiv \frac{PV}{RT} \Rightarrow \Delta Z^{lv} = \frac{P^{sat}\Delta V^{lv}}{RT} \Rightarrow \Delta V^{lv} = \frac{RT}{P^{sat}}\Delta Z^{lv}$$

ΔZ^{lv} = change in Z on vaporization (from roots of cubic EOS)

(P^{sat} and T are constant in phase change equilibrium)

$$\frac{dP^{sat}}{dT} = \frac{P^{sat}\Delta H^{lv}}{RT^2\Delta Z^{lv}}$$

$$\frac{dP^{sat}}{dT}\frac{1}{P^{sat}} = \frac{\Delta H^{lv}}{RT^2\Delta Z^{lv}}$$

$$\frac{1}{P^{sat}}dP^{sat} = d\ln P^{sat}$$

$$d(1/T) = -\frac{1}{T^2}dT$$

$$dT = -T^2 d(1/T)$$

$$\frac{d\ln P^{sat}}{dT} = \frac{\Delta H^{lv}}{RT^2\Delta Z^{lv}} \quad \text{(Eq. 6.87)}$$

$$\frac{d\ln P^{sat}}{d(1/T)} = -\frac{\Delta H^{lv}}{R\Delta Z^{lv}} \quad \text{(Eq. 6.88)}$$

T-Dependence of Vapor Pressure

(Clausius-Clapeyron)

$$\Delta H^{lv} = -R\frac{d\ln P^{sat}}{d(1/T)} \Rightarrow \frac{d\ln P^{sat}}{d(1/T)} = -\frac{\Delta H^{lv}}{R} = -B$$

(constant)

$$d\ln P^{sat} = -B \cdot d(1/T)$$

Integrate:

$$\ln P^{sat} = A - \frac{B}{T} \quad \text{(Eq. 6.89)}$$

- Plot of ln P^{sat} vs 1/T is straight line
- A and B are constants determined from regression.
- Valid from triple point to critical point

Louis Charles Antoine, 1825-1897
"Vapor Pressure: a new relationship between pressure and temperature, *Comptes Rendus de l'Academie des Sciences* (Proceedings of the French Academy of Science), 1888, and in *Annales de Physique et de Chimie,* 1891

(Eq. 6.90)

$$\ln P^{sat} = A - \frac{B}{T + C}$$

- **Antoine Equation**
- Improved version.
- Addition of constant "C"
- Constants in Table B.2

Total Properties from Δ's

Two-Phase Liquid/Vapor Systems

For any extensive property, such as total volume nV:

$$nV = n^{liquid}V^{liquid} + n^{vapor}V^{vapor} = n^{l}V^{l} + n^{v}V^{v}$$

n = total moles
- n^{l} = moles of liquid
- n^{v} = moles of vapor

V = molar volume
- V^{l} = molar volume of liquid
- V^{v} = molar volume of vapor

$$n = n^{l} + n^{v}$$

$$\frac{n}{n} = \frac{n^{l}}{n} + \frac{n^{v}}{n}$$

$$\frac{nV}{n} = \frac{n^{l}V^{l}}{n} + \frac{n^{v}V^{v}}{n}$$

$$1 = x^{l} + x^{v}$$

$$V = x^{l}V^{l} + x^{v}V^{v}$$

$$V = \left(1 - x^{v}\right)V^{l} + x^{v}V^{v}$$

$x^{v} \equiv$ quality of vapor

$$M = \left(1 - x^{v}\right)M^{l} + x^{v}M^{v} \qquad \text{(Eq. 6.96a)}$$

$M \equiv$ V, U, H, S, G

$$M = M^{l} + x^{v}\Delta M^{lv} \qquad \text{(Eq. 6.96b)}$$

(Important for upcoming lessons)

Questions?